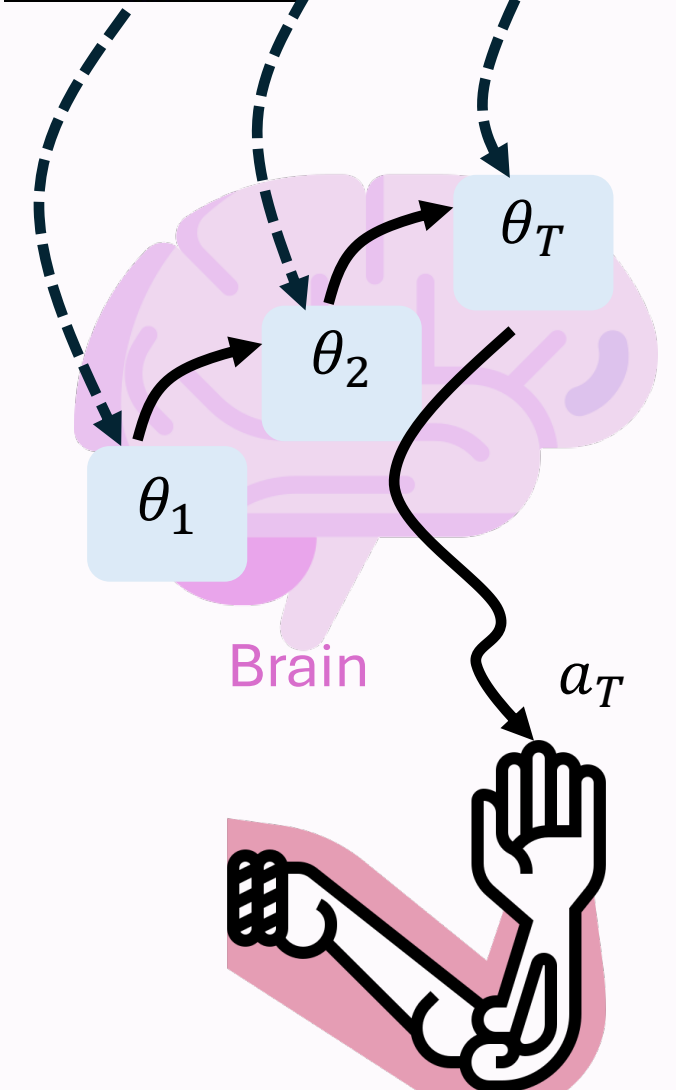


Biologically Human Behavior

Environment

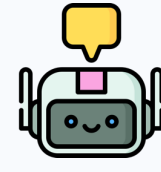
O_1 O_2 ... O_T



Human Action

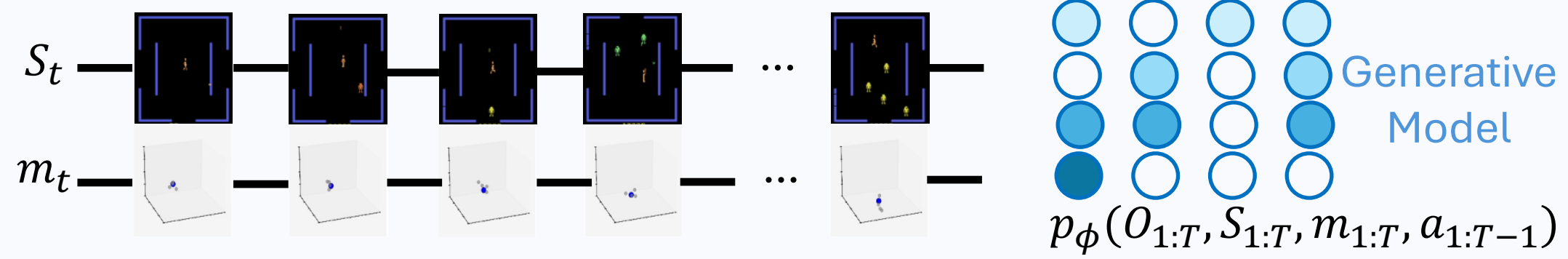
Query LLM for Intention

States m_t : Decide how many latent states should be used. Provide interpretation.....



.....

World Model



Habitual Policy and Action Selection

Habitual Rule Construction

Rule-1: not move forward ← an enemy is within one step directly

Rule-2: prioritize shooting ← an enemy is visible in the same row

Rule Activation

Rule- f is active at time t iff

$$\kappa_r(S_t^{MAP}, S_r) \geq \tau_r$$

$$m_t^{MAP} = m_r^*$$

Hybrid Policy

$$p_{\phi_\pi}(a_t | S_t, m_t, \mathcal{F})$$

Rule Hit?

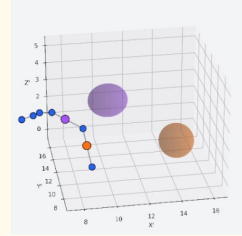
✓ Rule Action a_t

✗ EFE Planning (Beam Search)

Wake Phase

$$\min_{\phi, \vartheta} \mathbb{E}_{(O_t, a_t) \sim \mathcal{D}_{real}} \mathbb{E}_{q_{\vartheta}(S_t, m_t | \mathcal{H}_t)} [VFE(O_t, S_t, m_t; \phi, \vartheta)]$$

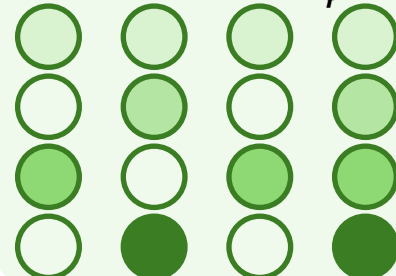
Create New Rules with Confidence



New Rule-1 ρ_1
New Rule-2 ρ_2

Sleep Phase

$$\min_{\phi, \phi_\pi} \mathbb{E}_{p_\phi(S, O, m, a)} [VFE(O, S, m; \phi, \vartheta) + \eta EFE(a; \phi, \phi_\pi) + \gamma D_{KL}(\cdot)]$$



Generate Replayed Trajectories



Refined Rule-1 ρ_1^*
Refined Rule-2 ρ_2^*

Goal: Minimize Free Energy

